

Supply Chain Resilience & Capital Allocation Engine

Rebuild Blueprint — From Acquired System to Target Architecture

1. Baseline: What We Acquired

Reference architecture: Current State — Forensic Audit v1.0.

The acquired system is a lightweight Streamlit application containing a substantive optimization and stochastic simulation foundation, interactive executive UI, supply-chain data ingestion, market-sector telemetry, reporting, authentication, and CI/CD infrastructure.

The current system flow is:

Data Ingestion → Model Build → Optimization → Simulation → Risk & Allocation Results → Reporting / UI → Decision Support

The strongest existing components are:

- Mixed-integer capital allocation optimization
- Semi-continuous allocation constraints
- Dependency constraints
- Bundle/synergy economics
- Target-risk optimization
- Correlated Monte Carlo simulation
- P50/P90 tail-risk analysis
- Interactive executive dashboard
- Supply-chain spreadsheet ingestion and validation
- 11-sector market telemetry
- Last-known market-state fallback
- Reporting/export capabilities
- Existing AWS/GitHub Actions deployment foundation

The current system map identifies optimization, Monte Carlo, UI/UX, and core data ingestion as the strongest existing domains.

2. What Is Valuable

A. Optimization Engine — HIGH VALUE

The system contains a real optimization foundation rather than merely displaying analytics.

The model incorporates:

- capital constraints
- risk-reduction objectives
- lead-time objectives
- semi-continuous investment variables
- prerequisite dependencies
- bundle discounts
- target-risk mode
- sensitivity/budget sweeps

This is the computational core that should be preserved, isolated, tested, and expanded.

Disposition: KEEP / STRENGTHEN

B. Stochastic Simulation — HIGH VALUE

The system already uses correlated stochastic variables and Cholesky factorization to generate simulated outcomes and P50/P90 risk measures.

This should become the foundation for the future stochastic engine rather than being discarded.

Disposition: KEEP / RE-ENGINEER / VALIDATE

C. 11-Sector Market Intelligence — STRATEGIC VALUE

The system currently tracks eleven stock-market sectors through sector ETFs and polls market information on a 30-minute cycle. It already contains sticky last-known-value behavior when feeds fail.

This is important because market closure or temporary feed failure should **not** be interpreted as a neutral market condition.

The target behavior is:

Last valid observation → persisted historical record → current state = last known value until superseded by a new observation

Every observation should retain:

- timestamp
- market/session status
- source
- sector
- price/value

- daily change
- volatility
- data quality/status

The market-data layer should become a persistent time-series subsystem rather than session-state storage.

Disposition: KEEP / MAJOR REFACTOR

3. What Is Not Yet Proven

Several capabilities are currently represented conceptually but require scientific or engineering validation.

Risk Model

The current model uses user-provided risk-reduction percentages as core inputs. The application explicitly distinguishes these from probabilities or financial-loss estimates.

Conclusion: this is currently a configurable resilience optimization model, not yet a complete probabilistic supply-chain digital twin.

Disposition: VALIDATE / EXPAND

Market Catalyst Engine

Current anomaly/catalyst detection is useful but preliminary.

Disposition: REBUILD AS A DEDICATED MARKET INTELLIGENCE SERVICE

Forecasting

Forward 5-day, 30-day, 90-day, and 1-year stochastic forecasting does not yet exist at the required level.

Disposition: NEW CORE CAPABILITY

AI Copilot

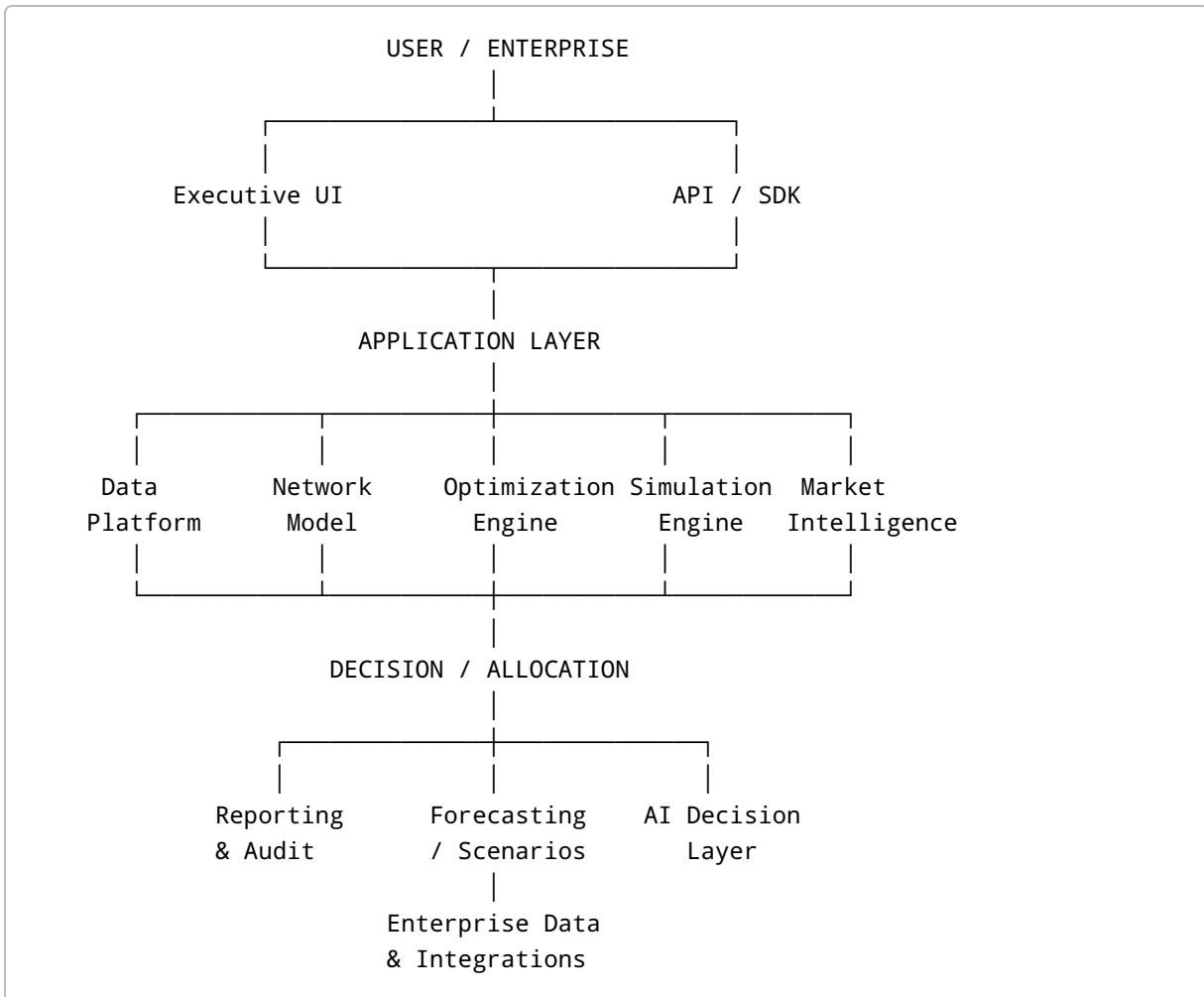
The current copilot is a deterministic command parser rather than a genuine reasoning/AI layer.

Disposition: KEEP INTERFACE CONCEPT / REBUILD INTELLIGENCE LAYER

4. What Must Be Rebuilt

The current Streamlit monolith should not become the final architecture.

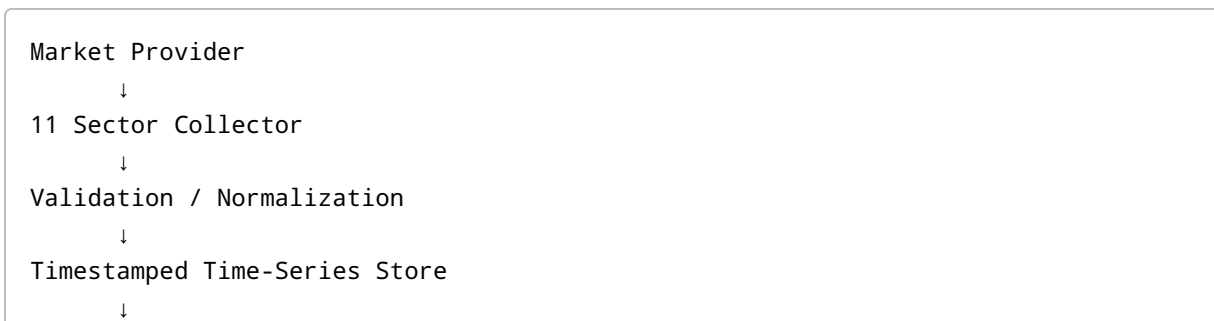
The target architecture is:



The UI should become a **client of the engine**, not the location where the engine lives.

5. Market Intelligence Target Architecture

The 11-sector system becomes its own persistent subsystem.



```
Latest-Known-State Cache
  ↓
Anomaly Detection
  ↓
Catalyst Detection
  ↓
Cross-Sector Correlation
  ↓
Stochastic Forecasting
  ↓
Scenario Generator
  ↓
Supply-Chain Risk / Capital Model
```

Forecast horizons

The forecasting engine will support:

- 5 days
- 30 days
- 90 days
- 1 year

Each horizon should produce distributions rather than false precision:

- expected state
- confidence/distribution bands
- tail scenarios
- probability of material movement
- sector relationships
- detected catalysts
- persistence/decay of catalysts

6. Stochastic Engine Vision

The long-term differentiator is not pretending that Black Swans can be predicted exactly.

The goal is to represent uncertainty more expressively.

The stochastic system should combine:

Observed Data

- **Historical Distributions**

- **Cross-Sector Correlations**
- **Volatility**
- **Regime Changes**
- **Known Catalysts**
- **External High-Entropy Inputs**
- **Supply-Chain Dependencies**

to produce a bounded distribution of plausible futures.

Entropy research track

The biological/biochip concept becomes an experimental stochastic-input layer.

It should not contaminate the deterministic production engine.

Instead:

Production mode → reproducible deterministic seeds

Operational stochastic mode → secure/high-quality entropy

Research mode → experimental external entropy sources, including biological signals if technically validated

This allows the original idea to become a legitimate research program without making the commercial engine dependent on an unproven mechanism.

7. Testing & Model Governance

This is one of the largest gaps in the acquired system.

The rebuilt product requires:

Unit tests

Every domain component.

Integration tests

Data → model → optimization → simulation.

Mathematical validation

Known-answer optimization problems.

Property-based testing

Constraints that must always hold.

Monte Carlo validation

Seed reproducibility, convergence, correlation integrity, percentile behavior, edge cases.

Performance testing

Network size and simulation-count benchmarks.

Regression suite

Every future engine modification must prove that previously validated scenarios still behave correctly.

Model versioning

Every result should identify:

- model version
- data version
- scenario version
- parameter set
- stochastic seed
- source timestamps

This becomes **model governance**, not merely software testing.

8. Security & Enterprise Architecture

The current authentication foundation is useful, but enterprise readiness requires substantially more.

Target capabilities:

- SSO
- SAML/OIDC
- RBAC
- tenant isolation
- row-level security
- encrypted secrets
- audit logging
- API authentication
- least privilege
- dependency scanning

- SAST/DAST
- container scanning
- infrastructure scanning
- data retention controls
- deployment isolation

9. CI/CD Rebuild

The existing GitHub workflows are valuable starting material.

Every workflow will be classified:

KEEP / MODIFY / REPLACE / REMOVE

The final pipeline should be:

```
Commit
↓
Format / Lint
↓
Type Check
↓
Unit Tests
↓
Integration Tests
↓
Model Validation
↓
Property Tests
↓
Security Scans
↓
Dependency Scan
↓
Performance Smoke Test
↓
Build
↓
Deploy
↓
Smoke Test
↓
Observability Verification
```

No production model change should bypass model validation.

10. Red / Yellow / Green Transition

GREEN — Preserve and strengthen

- Optimization
- Monte Carlo foundation
- UI/UX
- Data validation
- Reporting
- Core supply-chain modeling
- Market-sector concept
- Existing deployment foundation

YELLOW — Refactor and validate

- Market intelligence
- Persistent caching
- Supabase architecture
- Integrations
- AI copilot
- Scalability
- CI/CD
- Market anomaly/catalyst detection

RED — Rebuild

- Monolithic orchestration
- Testing architecture
- Security architecture
- Observability
- Forecasting
- Enterprise integrations
- Scientific model governance

BLUE — New capabilities

- 5d/30d/90d/1y stochastic forecasting
- Catalyst prediction
- Market regime detection
- Persistent market time series
- Scenario orchestration
- External entropy research layer
- Enterprise API
- ERP/WMS/TMS connectors

- Decision intelligence layer
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11. Final Target Product

The eventual product should not merely be described as a supply-chain dashboard.

The target definition is:

An enterprise decision engine that models supply-chain dependencies, allocates resilience capital, incorporates live market conditions, stress-tests correlated uncertainty, detects emerging catalysts, and generates probabilistic future scenarios across multiple time horizons.

The product's fundamental output becomes:

What should we fund?

How much should we fund it?

What happens if conditions change?

What risks remain?

What could happen next?

How confident are we?

Which external catalysts could change the answer?

That is the architecture and product thesis we will build toward from the acquired system.

12. Rebuild Principle

Do not rewrite blindly.

First preserve what has demonstrated value.

Then isolate it.

Then mathematically validate it.

Then test it.

Then improve it.

Then expand it.

The v1.0 system map remains the historical baseline against which every major architectural change can be compared.